

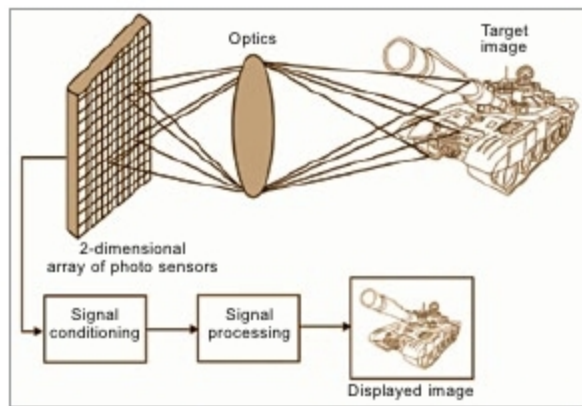


Fig. 6: Thermal imaging

electronic impulses into data for the display. Fig. 6 illustrates the concept of thermal imaging.

Types of thermal imaging sensors. There are two distinctive detector technologies: direct detection (or photon counting) and thermal detection. In direct detection, the detector element translates the photons directly into electrons. The charge accumulated, the current flow or the change in conductivity is proportional to the radiance of objects in the scene. Detectors in this category include lead selenide (PbSe), mercury cadmium telluride (HgCdTe), indium antimonide (InSb), platinum silicide (PtSi) and more.

All thermal imaging sensors based on direct detection technology, except those working in short-wave IR (SWIR), use detectors cooled to cryogenic temperatures close to -200°C . Newer photon-type IR sensors operating at elevated temperatures are now available. This has allowed solid-state thermal electric coolers and sterling coolers to be used.

Cooled thermal imagers are highly susceptible to damage from rugged use, have a long cooling time of typically a few minutes, limited MTBF of a few thousand hours, high cost, large size and weight, and high electrical power consumption leading to short battery life. The biggest advantage of these detectors is excellent spatial resolution and sensitivity that results from detector cooling.

Thermal detection, on the other hand, uses uncooled detectors. These make use of secondary effects such as relation between conductivity, capacitance and expansion, and detector temperature. Detectors in this category include bolometers, thermocouples, thermopiles and pyroelectric detectors. These sensors operate at room tempera-

ture and are lightweight. This feature, for example, allows microbolometer thermal imagers to be mounted on helmets.

Different generations of thermal imaging sensors. There are different generations of thermal imaging sensors. Each successive generation has incorporated not only a major change in the type of detector but also a major change in optical systems used to image the target onto the detector.

Four distinct generations of thermal imagers have been designed, based on IR detector technologies developed during the last 35 years, and classified according to the number of elements contained in each group. First-generation thermal imagers contain single element detectors, or detectors with only a few elements (1x3). A two-dimensional mechanical scanner was usually used to generate a two-dimensional image.

Sensitivity of thermal imaging sensors of the first generation was limited by background radiation. This problem was overcome in second-generation thermal imaging sensors by using modified front-end optics that reduced unwanted flux. However, this resulted in a fixed f-number for all fields-of-view.

Second-generation thermal imagers are vector detectors, usually containing 64 or more elements. The two-dimensional scanner was somehow simplified in the vertical direction to include only the interlace motion.

Third-generation thermal imagers contain dual-band, two-dimensional arrays with several columns of elements and dual/variable f-number optical system. These thermal imagers still scan in one direction and perform time-delay integration of the signal in the scanning direction to improve signal-to-noise ratio.

Fourth-generation thermal imagers contain two-dimensional array detectors (160x120, 320x240, 680x480) called focal plane arrays that do not require any scanning mechanism for acquiring two-dimensional pictures.

Applications of night vision equipment

Night vision equipment finds extensive usage in a wide range of appli-

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	A	0 - 2, 5
	V	0 - 20 0 - 200, 750 (3 Digits)
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